



PracticeTimeOnly

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הנושא שלנו הוא `TimeOnly` — טיפוס שמייצג שעה בלבד, בלי תאריך.

לפני התרגול: שלוש הפעולות החשובות

1. יצירת שעה מתוך טקסט

```
TimeOnly openAt = TimeOnly.Parse("8:00 AM");
```

`TimeOnly`. מקבל טקסט ומנסה להפוך אותו לערך מסוג `TimeOnly.Parse`.

2. קבלת השעה הנוכחית

```
TimeOnly rightNow = TimeOnly.FromDateTime(DateTime.Now);
```

כאן מתרחשים שני דברים:

`DateTime.Now`

מחזיר את התאריך והשעה הנוכחיים.

לאחר מכן:

```
TimeOnly.FromDateTime(...)
```

לוקח מתוך הערך הזה רק את השעה.

3. הדפסת שעה

```
Console.WriteLine(openAt);
```

תרגול TimeOnly

חלק א' — יצירת שעות

תרגיל 1 — שעת פתיחה

צרי משתנה מסוג `TimeOnly` בשם:

```
storeOpenAt
```

הכניסי לתוכו את השעה `AM 9:00` באמצעות `TimeOnly.Parse`, ולאחר מכן הדפיסי אותו.
התוצאה אמורה לייצג:

```
בבוקר 9:00
```

תרגיל 2 — שעת סגירה

צרי משתנה בשם:

```
storeCloseAt
```

שישמור את השעה `PM 8:00`.

הדפיסי למסך:

```
The store closes at: [השעה]
```

השתמשי ב־string interpolation.

תרגיל 3 — שתי שעות

צרי שני משתנים:

```
TimeOnly schoolStartsAt  
TimeOnly schoolEndsAt
```

הערכים שלהם יהיו:

```
8:15 AM  
1:30 PM
```

הדפיסי שתי שורות:

```
School starts at: ...  
School ends at: ...
```

חלק ב' — השעה הנוכחית

תרגיל 4 — מה השעה עכשיו?

צרי משתנה בשם:

```
currentTime
```

השתמשי ב:

```
TimeOnly.FromDateTime(DateTime.Now)
```

והדפיסי:

```
The current time is: ...
```

תרגיל 5 — הפרדה בין `DateTime` ל-`TimeOnly`

צרי משתנה מסוג `DateTime` בשם:

```
currentDateTime
```

והכניסי אליו:

```
DateTime.Now
```

לאחר מכן צרי משתנה מסוג `TimeOnly` בשם:

```
currentTime
```

וחלצי אליו רק את השעה מתוך `currentDateTime`.

הדפיסי את שני המשתנים ובדקי את ההבדל:

```
Full date and time: ...
```

```
Time only: ...
```

חלק ג' — עיצוב השעה

תרגיל 6 — הצגה בפורמט של 24 שעות

צרי שעה:

```
TimeOnly movieStartsAt = TimeOnly.Parse("7:45 PM");
```

הדפיסי אותה בפורמט:

```
19:45
```

רמז:

```
.ToString("HH:mm")
```

תרגיל 7 — הצגה בפורמט של 12 שעות

צרי שעה שמייצגת `6:30` בערב והדפיסי אותה כך:

```
06:30 PM
```

רמז:

```
.ToString("hh:mm tt")
```

שימי לב:

- `HH` — פורמט של 24 שעות.
- `hh` — פורמט של 12 שעות.
- `mm` — דקות.
- `tt` — `AM` או `PM` מציג.

חלק ד' — השוואת שעות

תרגיל 8 — לפני או אחרי?

צרי:

```
TimeOnly appointmentTime = TimeOnly.Parse("3:30 PM");  
TimeOnly currentTime = TimeOnly.FromDateTime(DateTime.Now);
```

בדקי:

- אם השעה הנוכחית קטנה משעת הפגישה, הדפיסי:

```
The appointment has not started yet.
```

- אחרת, הדפיסי:

```
The appointment time has arrived or passed.
```

מבנה התחלתי:

```
if (currentTime < appointmentTime)  
{  
  
}  
else  
{
```

```
}
```

תרגיל 9 — האם העסק פתוח?

הגדירי:

```
TimeOnly openAt = TimeOnly.Parse("8:00 AM");  
TimeOnly closeAt = TimeOnly.Parse("7:00 PM");  
TimeOnly currentTime = TimeOnly.FromDateTime(DateTime.Now);
```

בדקי אם השעה הנוכחית נמצאת בטווח שעות הפעילות.

כאשר העסק פתוח:

```
The business is open.
```

כאשר העסק סגור:

```
The business is closed.
```

התנאי צריך לבדוק:

```
currentTime >= openAt && currentTime < closeAt
```

חלק ה' — תרגול חשיבה

בתרגילים הבאים אינני נותן לך את מבנה הקוד. את צריכה לפרק בעצמך את הבעיה למשתנים, תנאים והדפסות.

תרגיל 10 — שעות פעילות של חדר כושר

חדר הכושר:

- נפתח בשעה 6:00 AM.
- נסגר בשעה 10:00 PM.

כתבי תוכנית שמציגה:

The gym is currently open.

או:

The gym is currently closed.

בהתאם לשעה הנוכחית.

תרגיל 11 — זמן יציאה לבית הספר

שעת היציאה לבית הספר היא:

7:30 AM

בדקי את השעה הנוכחית:

• לפני 7:30:

It is not time to leave yet.

• בדיוק ב-7:30 או אחריה:

It is time to leave.

תרגיל 12 — ברכה לפי השעה

השתמשי בשעה הנוכחית והדפיסי:

• לפני 12:00 PM :

Good morning!

• מ- 12:00 PM ועד לפני 6:00 PM :

Good afternoon!

• מ- 6:00 PM והלאה:

Good evening!



אתגר מסכם

מערכת שעות פתיחה לבית קפה

כתבי תוכנית עבור בית קפה:

Opening time: 7:00 AM

Closing time: 8:30 PM

התוכנית צריכה:

1. לשמור את שעת הפתיחה במשתנה מסוג `TimeOnly`.

2. לשמור את שעת הסגירה במשתנה מסוג `TimeOnly`.

3. לקבל את השעה הנוכחית.

4. להציג את כל השעות בפורמט `HH:mm`.

5. לבדוק אם בית הקפה פתוח.

6. להציג הודעה מתאימה.

דוגמה אפשרית לפלט:

Current time: 14:25

Opening time: 07:00

Closing time: 20:30

The coffee shop is open.

חוקי התרגול שלי אליך

- כתבי כל תרגיל בעצמך, גם כאשר נדמה לך שהפתרון ברור.
- אל תעתיקי את התרגיל הקודם ותשני רק את הערכים.
- הקלידי מחדש את שמות המשתנים ואת התחביר.
- הריצי את הקוד לאחר כל תרגיל.
- כאשר יש שגיאה, קראי קודם את הודעת השגיאה ונסי להבין אותה לפני תיקון אוטומטי.

המטרה איננה רק לדעת מה `TimeOnly` עושה — אלא לגרום לידיים שלך לדעת לכתוב אותו בלי לחשוב יותר מדי. 🧠💻

```
//`TimeOnly` Exercise
//Part 1 – Creating Hours
//Exercise 1 – Opening Time
//Create a `TimeOnly` variable named:
//storeOpenAt
// Input the time `9:00 AM` into it using `TimeOnly.Parse`,
// and then print it.
//The result should represent:
//9:00 AM

TimeOnly storeOpenAt = TimeOnly.Parse("9:00 AM");
string timeTypeMorning = storeOpenAt.ToString("h:mm tt");
Console.WriteLine(timeTypeMorning);
//~~~~~
~~~~~

//Exercise 2 – Closing Time
//Create a variable named:
//storeCloseAt
// that will hold the time `8:00 PM`.
// Print to the screen:
// The store closes at: [time]
// Use string interpolation.

TimeOnly storeCloseAt = TimeOnly.Parse("20:00");
string timeTypeEvening = storeCloseAt.ToString("h:mm tt");
Console.WriteLine($"The store closes at: {timeTypeEvening}");
//~~~~~
~~~~~

//Exercise 3 – Two Hours
//Create two variables:
//TimeOnly schoolStartsAt
//TimeOnly schoolEndsAt
```

```

//Their values will be:
//8:15 AM
// 1:30 PM
//Print two lines:
//School starts at: ...
//School ends at: ...

TimeOnly schoolStartsAt = TimeOnly.Parse("8:15");
string timeTypeMorning = schoolStartsAt.ToString("h:mm t
t");
Console.WriteLine($"School starts at: {timeTypeMorning}");

TimeOnly schoolEndsAt = TimeOnly.Parse("13:30");
string timeTypeEvening = schoolEndsAt.ToString("h:mm tt");
Console.WriteLine($"School ends at: {timeTypeEvening}");
//~~~~~
~~~~~

//Part 2 – The Current Time
//Exercise 4 – What time is it now?
//Create a variable named:
//currentTime
//Use:
//TimeOnly.FromDateTime(DateTime.Now)
//And print:
//The current time is: ...

TimeOnly currentTime = TimeOnly.FromDateTime(DateTime.Now);
Console.WriteLine($"The current time is: {currentTime}");
//~~~~~
~~~~~

//Exercise 5 – Differentiating `DateTime` from `TimeOnly`
//Create a `DateTime` variable named:
//currentDateAndTime
//and enter into it:
//DateTime.Now
//Then create a `TimeOnly` variable named:

```

```

//currentTime
//and extract only the time from `currentDateAndTime`.
//Print the two variables and check the difference:
//Full date and time: ...
//Time only: ...

DateTime currentDateAndTime = DateTime.Now;

TimeOnly currentTimeOnly = TimeOnly.FromDateTime(currentDateAndTime);

Console.WriteLine($"Full date and time: {currentDateAndTime}");
Console.WriteLine($"Time only: {currentTimeOnly}");
//~~~~~
~~~~~

//Part 3 – Formatting the Time
//Exercise 6 – Displaying in 24-Hour Format
//Create a time:
//TimeOnly movieStartsAt = TimeOnly.Parse("7:45 PM");
//Print it in the format:
//19:45
//Hint:
//.ToString("HH:mm")

TimeOnly movieStartsAt = TimeOnly.Parse("7:45 PM");
string movieTime = movieStartsAt.ToString("HH:mm");
Console.WriteLine(movieTime);
//~~~~~
~~~~~

//Exercise 7 – 12-hour format
//Create a time that represents `6:30` in the evening and print it like this:
//06:30 PM
//Hint:
//.ToString("hh:mm tt")

```

```

//Note:
//`HH` - 24-hour format.
//`hh` - 12-hour format.
//`mm` - minutes.
//`tt` - displays `AM` or `PM`.

TimeOnly eveningTime = TimeOnly.Parse("6:30 PM");
string sixtyPm = eveningTime.ToString("hh:mm tt");
Console.WriteLine(sixtyPm);
//~~~~~
~~~~~

//Part 4 - Comparing Times
//Exercise 8 - Before or After?
//Create:
//TimeOnly appointmentTime = TimeOnly.Parse("3:30 PM");
//TimeOnly currentTime = TimeOnly.FromDateTime(DateTime.No
w);
//Check:
//If the current time is less than the appointment time, pr
int:
//The appointment has not started yet.
//Otherwise, print:
//The appointment time has arrived or passed.
//Initial structure:
//if (currentTime < appointmentTime)
//{
//}
//else
//{
//}

TimeOnly appointmentTime = TimeOnly.Parse("3:30 PM");
TimeOnly currentTime2 = TimeOnly.FromDateTime(DateTime.No
w);

if (currentTime2 < appointmentTime)
{

```

```

        Console.WriteLine("The appointment has not started yet");
    }
    else
    {
        Console.WriteLine("The appointment time has arrived or passed");
    }
    //~~~~~
    ~~~~~

//Exercise 9 – Is the business open?
//Define:
//TimeOnly openAt = TimeOnly.Parse("8:00 AM");
//TimeOnly closeAt = TimeOnly.Parse("7:00 PM");
//TimeOnly currentTime = TimeOnly.FromDateTime(DateTime.Now);
//Check if the current time is within the operating hours.
//When the business is open:
//The business is open.
//When the business is closed:
//The business is closed.
//The condition should be checked:
//currentTime >= openAt && currentTime < closeAt

TimeOnly openAt = TimeOnly.Parse("8:00 AM");
TimeOnly closeAt = TimeOnly.Parse("7:00 PM");
TimeOnly currentTime3 = TimeOnly.FromDateTime(DateTime.Now);

if (currentTime3 >= openAt && currentTime3 < closeAt)
{
    Console.WriteLine("The business is open");
}
else
{
    Console.WriteLine("The business is closed");
}

```

```
//~~~~~

//Exercise 10 – Gym Hours
//The gym:
//Opens at `6:00 AM`.
//Closes at `10:00 PM`.
//Write a program that displays:
//The gym is currently open.
//or:
//The gym is currently closed.
//depending on the current time.

TimeOnly opensAt = TimeOnly.Parse("6:00 AM");
TimeOnly closesAt = TimeOnly.Parse("10:00 PM");
TimeOnly currentTime4 = TimeOnly.FromDateTime(DateTime.Now);

if (currentTime4 >= opensAt && currentTime4 < closesAt)
{
    Console.WriteLine("The gym is currently open.");
}
else
{
    Console.WriteLine("The gym is currently closed.");
}
//~~~~~

//Exercise 11 – School Leaving Time
//The school leaving time is:
//7:30 AM
//Check the current time:
//Before 7:30:
//It is not time to leave yet.
//Exactly at or after 7:30:
//It is time to leave.
```

```

TimeOnly schoolLeaving = TimeOnly.Parse("7:30 AM");
string sevenThirtyAm = schoolLeaving.ToString("h:mm tt");
Console.WriteLine(sevenThirtyAm);

TimeOnly currentTime5 = TimeOnly.FromDateTime(DateTime.No
w);

if (currentTime5 < schoolLeaving)
{
    Console.WriteLine("It is not time to leave yet");
}
else
{
    Console.WriteLine("It is time to leave");
}
//~~~~~
~~~~~

//Exercise 12 – Greeting by Time
//Use the current time and print:
//Before `12:00 PM`:
//Good morning!
//From `12:00 PM` to before `6:00 PM`:
//Good afternoon!
//From `6:00 PM` onwards:
//Good evening!

TimeOnly currentTime6 = TimeOnly.FromDateTime(DateTime.No
w);

TimeOnly noon = TimeOnly.Parse("12:00 PM");
TimeOnly evening = TimeOnly.Parse("6:00 PM");

if (currentTime6 < noon)
{
    Console.WriteLine("Good morning!");
}
else if (currentTime6 < evening)

```

```

{
    Console.WriteLine("Good afternoon!");
}
else
{
    Console.WriteLine("Good evening!");
}
//~~~~~
~~~~~

//Final Challenge 🏆
//Opening hours system for a coffee shop
// Write a program for a coffee shop:
//Opening time: 7:00 AM
//Closing time: 8:30 PM

//The program should:
// 1. Store the opening time in a `TimeOnly` variable.
// 2. Store the closing time in a `TimeOnly` variable.
// 3. Get the current time.
// 4. Display all hours in `HH:mm` format.
// 5. Check if the coffee shop is open.
// 6. Display an appropriate message.
//Possible output example:
//Current time: 14:25
//Opening time: 07:00
//Closing time: 20:30
//The coffee shop is open.

TimeOnly openingTime = TimeOnly.Parse("7:00 AM");
string sevenAm = openingTime.ToString("HH:mm");
Console.WriteLine($"Opening time: {sevenAm}");

TimeOnly closingTime = TimeOnly.Parse("8:30 PM");
string eightAndAHalf = closingTime.ToString("HH:mm");
Console.WriteLine($"Closing time: {eightAndAHalf}");

TimeOnly currentTime7 = TimeOnly.FromDateTime(DateTime.No

```



```
w);
string timeNow = currentTime7.ToString("HH:mm");
Console.WriteLine($"Current time: {timeNow}");

if (currentTime7 >= openingTime && currentTime7 < closingTime)
{
    Console.WriteLine("The coffee shop is open.");
}
else
{
    Console.WriteLine("The coffee shop is closed.");
}
//~~~~~
~~~~~
```

תרגול מקיף: **TimeOnly**

המטרה: להרגיש ש- **TimeOnly** הוא "רעיון של שעה" בלי תאריך, ולדעת ליצור/להדפיס/לעצב/להשוות שעות — במיוחד בהקשר של שעות פתיחה של עסק מקומי.

חלק 1 — יצירה והדפסה (הבסיס)

תרגיל 1 — יצירת שעת פתיחה

1. צרי משתנה בשם **openAt** מסוג **TimeOnly**.
2. תני לו ערך של **AM 8:00** בעזרת **TimeOnly.Parse**.
3. הדפיסי אותו.

תרגיל 2 — יצירת שעת סגירה

1. צרי משתנה בשם **closeAt** עם השעה **PM 7:00**.
2. הדפיסי:

Closing time: ...

חלק 2 — "רק השעה" מתוך **DateTime**

תרגיל 3 — השעה הנוכחית

1. צרי משתנה בשם `rightNow` מסוג `TimeOnly` בעזרת

```
.TimeOnly.FromDateTime(DateTime.Now)
```

2. הדפיסי:

- את `DateTime.Now`

- את `rightNow`

מטרה: לראות ש-`DateTime` מכיל גם תאריך וגם שעה, ו-`TimeOnly` מכיל רק שעה.

תרגיל 4 — הוכחה ש-**TimeOnly** לא יודע תאריך

נסי לכתוב קוד שמנסה להציג "תאריך" מתוך `TimeOnly` (למשל חודש/יום).

המטרה: להבין לבד למה זה לא אפשרי — כי אין תאריך.

חלק 3 — עיצוב הצגה (Formatting)

תרגיל 5 — פורמט 24 שעות

1. צרי שעה בשם `meetingAt` בעזרת `TimeOnly.Parse` (בחרי שעה משלך, למשל שעה בערב עם דקות).

2. הדפיסי אותה בפורמט 24 שעות (HH:mm).

תרגיל 6 — פורמט 12 שעות

הדפיסי את אותה שעה בפורמט:

06:05 PM

חלק 4 — השוואות שעות

תרגיל 7 — האם כבר פתחנו?

כתבי תנאי בעזרת `openAt` ו- `rightNow` :

• אם `rightNow` לפני `We are not open yet` → `openAt` .

• אחרת → `We are open or already past opening time` .

תרגיל 8 — האם העסק פתוח (טווח שעות)

הגדירי שעות פתיחה וסגירה (`openAt/closeAt`) וקחי את השעה הנוכחית (`rightNow`).

בדקי האם `rightNow` נמצא בתוך הטווח של שעות הפעילות, והדפיסי:

- `The business is open.`
- `The business is closed.`

חלק 5 — תרגול "עסק מקומי" + חריגים

תרגיל 9 — חריג: סגירה מוקדמת

תסריט:

• בדרך כלל: פתיחה 8:00, סגירה 19:00

• היום יש חריג: סגירה מוקדמת ב-16:00

כתבי קוד שמגדיר:

- `regularCloseAt` (19:00)
- `earlyCloseAt` (16:00)
- `rightNow`

ואז בחרי "סגירה בפועל":

• אם היום חריג → השתמשי ב- `earlyCloseAt`

• אחרת → השתמשי ב- `regularCloseAt`

אתגר מסכם — מערכת שעות פתיחה חכמה (רק עם TimeOnly)

כתבי תוכנית שמדפיסה:

1. Right now: HH:mm
2. Open at: HH:mm
3. Close at: HH:mm

4. מצב: OPEN / CLOSED

5. ואם סגור — תדפיסי גם למה:

- Before opening
- After closing

```
//Part 1 – Creating and Printing (The Basics)
//Exercise 1 – Creating an Opening Time
//1. Create a variable:
//TimeOnly openAt;
//2. Give it a value of `8:00 AM` using `TimeOnly.Parse`.
// 3. Print it.

TimeOnly openAt = TimeOnly.Parse("8:00 AM");
string eightAm = openAt.ToString("h:mm tt");
Console.WriteLine(eightAm);
//~~~~~
~~~~~

//Exercise 2 – Creating a closing time
//1. Create a variable `closeAt` with the time `7:00 PM`.
//2. Print:
//Closing time: ...

TimeOnly closeAt = TimeOnly.Parse("7:00 PM");
string sevenPm = closeAt.ToString("h:mm tt");
Console.WriteLine($"Closing time: {sevenPm}");
//~~~~~
~~~~~
```

```
//Part 2 – “Just the Time” from `DateTime`
//Exercise 3 – The Current Time
//1. Create a variable named `rightNow` of type `TimeOnly`
using `TimeOnly.FromDateTime(DateTime.Now)` .
//2. Print:
//`DateTime.Now`
//`rightNow`
//Goal: To see that `DateTime` contains both a date and a t
ime, and `TimeOnly` contains only a time.
```

```
TimeOnly rightNow = TimeOnly.FromDateTime(DateTime.Now);
DateTime dateAndTime = DateTime.Now;
```

```
Console.WriteLine(rightNow);
Console.WriteLine(dateAndTime);
//~~~~~
~~~~~
```

```
//Part 3 – Formatting
//Exercise 5 – 24-hour format
//1. Create a time named `meetingAt` using `TimeOnly.Parse`
(choose your own time, e.g. PM with minutes).
//2. Print it in 24-hour format (HH:mm).
```

```
TimeOnly meetingAt = TimeOnly.Parse("12:00");
string withMinutes = meetingAt.ToString("HH:mm");
Console.WriteLine(withMinutes);
//~~~~~
~~~~~
```

```
//Exercise 6 – 12-hour format
//Print the same time in the format:
//06:05 PM
```

```
TimeOnly meetingAt2 = TimeOnly.Parse("06:05 PM");
string withMinutes2 = meetingAt2.ToString("hh:mm tt");
Console.WriteLine(withMinutes2);
```

```
//~~~~~

//Part 4 – Time Comparisons
//Exercise 7 – Are We Open Yet?
//Conditionals Using `openAt` and `rightNow`:

//If `rightNow` Before `openAt` → `We are not open yet.`
//Else → `We are open or already past opening time.`

TimeOnly openAt2 = TimeOnly.Parse("8:00 AM");

TimeOnly rightNow2 = TimeOnly.Parse(DateTime.Now.ToString(
    "HH:mm"));

if (rightNow2 < openAt2)
{
    Console.WriteLine("We are not open yet.");
}
else
{
    Console.WriteLine("We are open or already past opening
time.");
}
//~~~~~

//Exercise 8 – Is the business open (hour range)
//Define opening and closing hours (openAt/closeAt) and get
the current time (rightNow).
//Check if `rightNow` is within the range of operating hour
s, and print:
//`The business is open.`
//`The business is closed.`

TimeOnly openAt3 = TimeOnly.Parse("08:00");
string eightAm24 = openAt3.ToString("HH:mm");
Console.WriteLine(eightAm24);
```

```

TimeOnly closeAt3 = TimeOnly.Parse("23:00");
string twentyThreePm = closeAt3.ToString("HH:mm");
Console.WriteLine(twentyThreePm);

TimeOnly rightNow3 = TimeOnly.FromDateTime(DateTime.Now);
string timeNow3 = rightNow3.ToString("HH:mm");
Console.WriteLine(timeNow3);

if (rightNow3 >= openAt3 && rightNow3 < closeAt3)
{
    Console.WriteLine("The business is open.");
}
else
{
    Console.WriteLine("The business is closed.");
}
//~~~~~
~~~~~

//Part 5 – “Local Business” Practice + Exceptions
//Exercise 9 – Exception: Early Close
//Script:
//Normally: Open 8:00, Close 19:00
//Today there is an exception: Early Close at 16:00
//Write code that defines:
//`regularCloseAt` (19:00)
//`earlyCloseAt` (16:00)
//`rightNow`
//Then select “Actual Close”:
//If today is an exception → use `earlyCloseAt`
//Otherwise → use `regularCloseAt`

TimeOnly regularCloseAt = TimeOnly.Parse("19:00");
TimeOnly earlyCloseAt = TimeOnly.Parse("16:00");

TimeOnly rightNow4 = TimeOnly.FromDateTime(DateTime.Now);

```

```

bool isExceptionToday = true;

TimeOnly actualCloseAt;

if (isExceptionToday)
{
    actualCloseAt = earlyCloseAt;
}
else
{
    actualCloseAt = regularCloseAt;
}

Console.WriteLine($"Right now: {rightNow4.ToString("HH:mm")});
Console.WriteLine($"Actual closing time: {actualCloseAt.ToString("HH:mm")});
//~~~~~
~~~~~

//You are building a small program that simulates a business's opening hours system.
//It has three times that are important to you:
//Current time
//Business opening time
//Business closing time
//Then the program needs to figure out what the business's status is based on the current time.

//1. Right now: HH:mm
//The program needs to display the current time.
//For example only:
//Right now: 14:25
//HH:mm means the time should be displayed in 24-hour format.

//2. Open at: HH:mm

```



```

//You need to decide on the business's opening time and save it.
//For example, if the business opens in the morning, the program will display the opening time in the requested format:
//Open at: ...

// 3. Close at: HH:mm
//The same goes for the closing time.
//The program needs to know when the business closed and display:
//Close at: ...

//4.Business status: OPEN or CLOSED
//Now comes the part where the program needs to make a decision.
//It needs to look at:
//The time now
//Opening time
//Closing time
//And decide:
//OPEN
//or:
//CLOSED
//That is, the question you need to answer in the program is:
//Should the business be open at the current time?
//5. If the business is closed – explain why
//This is the part that makes the challenge a "smart system."

//It is not enough to write:
//CLOSED
//If the business is closed, the program also needs to understand why it is closed.

//There are only two possible reasons in the exercise:
//The business has not yet opened

```

```

//That is, the current time is too early.

// Then it should appear:
//Before opening
//The business has already closed
//That is, the operating hours have already ended.
//Then it should appear:
//After closing
//So there are actually three possible situations
//From a conceptual point of view only:
//Situation A: Still too early → The business is closed because it has not yet opened.
//Situation B: We are within operating hours → The business is open.
//Situation C: It's too late → The business is closed because it has already closed.

TimeOnly currentTime = TimeOnly.FromDateTime(DateTime.Now);
string timeNow = currentTime.ToString("HH:mm");
Console.WriteLine($"Right now: {timeNow}");

TimeOnly openingTime = TimeOnly.Parse("06:00");
string six = openingTime.ToString("HH:mm");
Console.WriteLine($"Open at: {six}");

TimeOnly closingTime = TimeOnly.Parse("23:00");
string twentyThree = closingTime.ToString("HH:mm");
Console.WriteLine($"Close at: {twentyThree}");

if (currentTime < openingTime)
{
    Console.WriteLine("CLOSED");
    Console.WriteLine("Before opening");
}
else if (currentTime < closingTime)
{
    Console.WriteLine("OPEN");
}

```

```
else
{
    Console.WriteLine("CLOSED");
    Console.WriteLine("After closing");
}
```

🎯 TimeOnly — Weak Points Practice

התרגול בנוי במיוחד סביב ארבע נקודות שצריך לחזק אצלך:

מה אנחנו מחזקים	נקודת חולשה
נכון Formatting	tt ו- hh לעומת HH
גבולות של שעות	< , > , <= , >=
תנאים סותרים/מיותרים	לעומת &&
בניית תנאי פשוט ונכון	בניית תנאי פשוט ונכון

Part 1 — Time Formatting

Exercise 1 — 24-Hour Format

צרי משתנה מסוג `TimeOnly` בשם `departureTime` ושמרי בו את הערך `PM 8:45`.
הדפיסי אותו כך שהתוצאה תהיה בדיוק `20:45`.

Exercise 2 — 12-Hour Format

צרי משתנה בשם `meetingTime` מסוג `TimeOnly` ושמרי בו את הערך `PM 6:05`.
הדפיסי אותו כך שהתוצאה תהיה בדיוק `PM 06:05`.

Exercise 3 — Two Different Formats

צרי `TimeOnly` אחד בלבד בשם `movieTime` עם הערך `PM 9:30`.
מאותו משתנה הדפיסי שתי שורות (בדיוק בפורמט הבא):

- 24-hour format: 21:30
- 12-hour format: 09:30 PM

Part 2 — Simple Comparisons

Exercise 4 — Before the Appointment

צרי:

- `TimeOnly appointmentTime` (שמרי בו `4:00 PM`)

- `TimeOnly currentTime` (השעה הנוכחית)

אם השעה הנוכחית לפני שעת הפגישה:

- הדפיסי: `.The appointment is later`

אחרת:

- הדפיסי: `.The appointment time has arrived or passed`

Exercise 5 — Exact Boundary

חנוות נפתחת בשעה `AM 9:00`. בדקי את השעה הנוכחית.

אם השעה הנוכחית היא **9:00 בדיוק** או מאוחר יותר:

- הדפיסי: `.The store has opened`

אם היא לפני 9:00:

- הדפיסי: `.The store has not opened yet`

Part 3 — Is the Time Inside the Range?

Exercise 6 — Library Opening Hours

הספרייה:

- Opens: `08:00`
- Closes: `17:00`

היא פתוחה מ-**08:00 כולל** ועד לפני **17:00**.

הדפיסי אחת מהאפשרויות:

- `The library is open.`
- `The library is closed.`

Exercise 7 — Swimming Pool

הבריכה:

- Opens: `06:30 AM`
- Closes: `9:00 PM`

פתוחה בדיוק בשעת הפתיחה, וסגורה בדיוק בשעת הסגירה.

הדפיסי:

- `The swimming pool is open.`
- `.The swimming pool is closed` או

Exercise 8 — Computer Lab

מעבדת המחשבים פתוחה בין **AM 10:00** ל- **PM 4:30**.

השתמשי בשלושה משתני **TimeOnly**:

- **opensAt**
- **closesAt**
- **currentTime**

והדפיסי:

- **Computer lab: OPEN**

• או **Computer lab: CLOSED**

Part 4 — Avoid Contradictory Conditions

Exercise 9 — Leaving for Work

צריך לצאת לעבודה בשעה **AM 7:45**.

לפני הזמן:

- **Stay home.**

בדיוק בזמן או אחרי:

- **Leave for work.**

Exercise 10 — Closing Time

מסעדה נסגרת בשעה **PM 11:00**.

לפני הזמן:

- **The restaurant has not closed yet.**

בדיוק בזמן או אחרי:

- **The restaurant is closed.**

Part 5 — Three Time Ranges

Exercise 11 — Parts of the Day

הדפיסי אחת בלבד:

• לפני **PM** → **Morning 12:00**

• מ- **PM 12:00** ועד לפני **Afternoon 5:00** → **PM**

- מִ־ PM 5:00 והלאה → Evening

Exercise 12 — Store Message

הדפיסי הודעה אחת בלבד בהתאם לשעה:

- לפני 9:00 → The store has not opened yet . AM
- מִ־ AM 9:00 ועד לפני 8:00 → The store is open . PM
- מִ־ PM 8:00 והלאה → .The store is closed

Part 6 — Formatting + Logic Together

Exercise 13 — Pharmacy

בית המרקחת:

- Opening time: 08:30 AM
- Closing time: 07:30 PM

התוכנית צריכה להדפיס (בפורמט HH:mm):

- Current time: ...
- Opening time: ...
- Closing time: ...

ואז להדפיס:

- The pharmacy is open.
- או The pharmacy is closed.

🏆 Final Challenge — TimeOnly Mastery

בית קפה עובד בין AM 06:45 לבין PM 09:15 .

הדפיסי את שלוש השעות בפורמט HH:mm , בדקי אם עכשיו בתוך טווח הפעילות, והדפיסי אחת מהאפשרויות:

- The coffee shop is currently open.
- The coffee shop is currently closed.

```
//TimeOnly – Weak Points Practice
//HH vs. hh and tt Correct formatting
//<, >, <=, >= Hour boundaries
//&& vs ||
```

```
//Conflicting/redundant conditions Building a simple and correct condition
//Part 1 – Time Formatting
//Exercise 1 – 24-Hour Format

//Create a TimeOnly variable named:
//departureTime
//And put into it:
//8:45 PM

//Print it exactly like this:
// 20:45

TimeOnly departureTime = TimeOnly.Parse("08:45 PM");
string twentyFortyFive = departureTime.ToString("HH:mm");
Console.WriteLine(twentyFortyFive);
//~~~~~
~~~~~
```

```
//Exercise 2 – 12-Hour Format

//Create a variable named:
//meetingTime

//Save it:
//6:05 PM

//Print it:
//06:05 PM

TimeOnly meetingTime = TimeOnly.Parse("6:05 PM");
string sixOoFive = meetingTime.ToString("hh:mm tt");
Console.WriteLine(sixOoFive);
//~~~~~
~~~~~
```

```
//Exercise 3 – Two Different Formats

//Create only one TimeOnly named:
//movieTime

//With the value:
//9:30 PM

//From the same variable Print two lines:
// 24-hour format: 21:30
// 12-hour format: 09:30 PM

TimeOnly movieTime = TimeOnly.Parse("9:30 PM");

string twentyFourHourFormat = movieTime.ToString("HH:mm");
string twelveHourFormat = movieTime.ToString("hh:mm tt");

Console.WriteLine($"24-hour format: {twentyFourHourFormat}");
Console.WriteLine($"12-hour format: {twelveHourFormat}");
//~~~~~
~~~~~
```

```
//Part 2 – Simple Comparisons
//Exercise 4 – Before the Appointment

//Create:
//TimeOnly appointmentTime
//TimeOnly currentTime

//Appointment time:
//4:00 PM

//currentTime should contain the current time using the method you learned in class.

// If the current time is before the appointment time:
```



```
//The appointment is later.

//Otherwise:
//The appointment time has arrived or passed.

TimeOnly appointmentTime = TimeOnly.Parse("4:00 PM");
string forPm = appointmentTime.ToString("h:mm tt");
Console.WriteLine(forPm);

TimeOnly currentTime = TimeOnly.FromDateTime(DateTime.Now);
string timeNow = currentTime.ToString("HH:mm");
Console.WriteLine(timeNow);

if (currentTime < appointmentTime)
{
    Console.WriteLine("The appointment is later.");
}
else
{
    Console.WriteLine("The appointment time has arrived or
passed.");
}
//~~~~~
~~~~~
```

```
//Exercise 5 – Exact Boundary
//The store opens at:
//9:00 AM

//Check the current time.

//If the current time is exactly 9:00 AM or later:
//The store has opened.

//If it is before 9:00 AM:
//The store has not opened yet.

//Pay special attention to the case where the time is exact
```

ly 9:00 AM.

```
TimeOnly opensAt = TimeOnly.Parse("9:00 AM");  
string nineAm = opensAt.ToString("h:mm tt");  
Console.WriteLine(nineAm);
```

```
TimeOnly currentTime = TimeOnly.FromDateTime(DateTime.Now);  
string timeNow = currentTime.ToString("HH:mm");  
Console.WriteLine(timeNow);
```

```
if (currentTime >= opensAt)  
{  
    Console.WriteLine("The store has opened.");  
}  
else  
{  
    Console.WriteLine("The store has not opened.");  
}
```

```
//~~~~~  
~~~~~
```

```
//Part 3 – Is the Time Inside the Range?  
//Exercise 6 – Library Opening Hours
```

```
//The library:  
//Opens: 08:00
```

```
//Closes: 17:00
```

```
//It is considered open:  
//Starting exactly 08:00
```

```
//Until before 17:00
```

```
//The program should display:  
//The library is open.
```

```
//Or:
```

```

//The library is closed.
//Depending on the current time.

TimeOnly libraryOpens = TimeOnly.Parse("08:00");
string eightAm = libraryOpens.ToString("HH:mm");
Console.WriteLine(eightAm);

TimeOnly libraryClose = TimeOnly.Parse("17:00");
string seventeenPm = libraryClose.ToString("HH:mm");
Console.WriteLine(seventeenPm);

TimeOnly currentTime = TimeOnly.FromDateTime(DateTime.Now);
string timeNow = currentTime.ToString("HH:mm");
Console.WriteLine(timeNow);

if (currentTime >= libraryOpens && currentTime < libraryClose)
{
    Console.WriteLine("The library is open.");
}
else
{
    Console.WriteLine("The library is close.");
}
//~~~~~
~~~~~

```

```

//Exercise 7 – Swimming Pool

//The pool:
//Opens: 06:30 AM
//Closes: 9:00 PM

//Write a program that checks if it is open now.

//At exactly the opening time it is open.
//At exactly the closing time it is already closed.

```

```

//Output:
//The swimming pool is open.

//Or:
//The swimming pool is closed.

TimeOnly opens = TimeOnly.Parse("06:30 AM");
string sixAm = opens.ToString("HH:mm tt");
Console.WriteLine(sixAm);

TimeOnly closes = TimeOnly.Parse("9:00 PM");
string ninePm = closes.ToString("h:mm tt");
Console.WriteLine(ninePm);

TimeOnly currentTime = TimeOnly.FromDateTime(DateTime.Now);
string timeNow = currentTime.ToString("HH:mm tt");
Console.WriteLine(timeNow);

if (currentTime >= opens && currentTime < closes)
{
    Console.WriteLine("The swimming pool is open.");
}
else
{
    Console.WriteLine("The swimming pool is closed.");
}
//~~~~~
~~~~~

```

```

//Exercise 8 – Computer Lab

//The computer lab is open:
//10:00 AM - 4:30 PM

//Use three TimeOnly variables:
//opensAt
//closesAt

```

```

//currentTime

//and print:
//Computer lab: OPEN
//or:
//Computer lab: CLOSED

TimeOnly opensAt = TimeOnly.Parse("10:00 AM");
string tenAm = opensAt.ToString("HH:mm tt");
Console.WriteLine(tenAm);

TimeOnly closesAt = TimeOnly.Parse("4:30 PM");
string forPm = closesAt.ToString("h:mm tt");
Console.WriteLine(forPm);

TimeOnly currentTime = TimeOnly.FromDateTime(DateTime.Now);
string timeNow = currentTime.ToString("HH:mm tt");
Console.WriteLine(timeNow);

if (currentTime >= opensAt && currentTime < closesAt)
{
    Console.WriteLine("Computer lab: OPEN");
}
else
{
    Console.WriteLine("Computer lab: CLOSE");
}
//~~~~~
~~~~~

//Part 4 – Avoid Contradictory Conditions
//Exercise 9 – Leaving for Work

//You need to leave for work at:
//7:45 AM

//Before this time:
//Stay home.

```

```

//Exactly at 7:45 or after:
//Leave for work.

//There are only two situations here.

TimeOnly leaveForWork = TimeOnly.Parse("7:45 AM");
string sevenFortyFiveAm = leaveForWork.ToString("h:mm tt");
Console.WriteLine(sevenFortyFiveAm);

TimeOnly currentTime = TimeOnly.FromDateTime(DateTime.Now);
string timeNow = currentTime.ToString("HH:mm");
Console.WriteLine(timeNow);

if (currentTime >= leaveForWork)
{
    Console.WriteLine("Leave for work.");
}
else
{
    Console.WriteLine("Stay home.");
}
//~~~~~
~~~~~

//Exercise 10 – Closing Time

//A restaurant closes at:
//11:00 PM

//If it is still before closing time:
//The restaurant has not closed yet.

//Exactly at or after 11:00 PM:
//The restaurant is closed.

TimeOnly closeAt = TimeOnly.Parse("11:00 PM");
string elevenPm = closeAt.ToString("hh:mm tt");

```

```

Console.WriteLine(elevenPm);

TimeOnly currentTime = TimeOnly.FromDateTime(DateTime.Now);
string timeNow = currentTime.ToString("HH:mm");
Console.WriteLine(timeNow);

if (currentTime < closeAt)
{
    Console.WriteLine("The restaurant has not closed yet.");
}
else
{
    Console.WriteLine("The restaurant is closed.");
}
//~~~~~

//Part 5 – Three Time Ranges

//Exercise 11 – Parts of the Day

//Use the current time.

//The ranges are:
//Before 12:00 PM
//Morning
//12:00 PM until before 5:00 PM
//Afternoon
//5:00 PM and later
//Evening

//Print only one of the options:
//Morning
//Afternoon
//Evening

```

```

TimeOnly currentTime = TimeOnly.FromDateTime(DateTime.Now);
string timeNow = currentTime.ToString("HH:mm tt");
Console.WriteLine(timeNow);

TimeOnly morning = TimeOnly.Parse("12:00 PM");
string twelvePm = morning.ToString("HH:mm tt");
Console.WriteLine(twelvePm);

TimeOnly afternoon = TimeOnly.Parse("5:00 PM");
string fivePm = afternoon.ToString("h:mm tt");
Console.WriteLine(fivePm);

if (currentTime < morning)
{
    Console.WriteLine("Morning");
}
else if (currentTime >= morning && currentTime < afternoon)
{
    Console.WriteLine("Afternoon");
}
else
{
    Console.WriteLine("Evening");
}
//~~~~~
~~~~~

//Exercise 12 – Store Message

//A store operates according to the following rules:

//Before 9:00 AM
//The store has not opened yet.

//9:00 AM until before 8:00 PM
//The store is open.

//8:00 PM and later

```



```

//The store is closed.

//Use TimeOnly only for business hours.

TimeOnly open = TimeOnly.Parse("9:00 AM");
string nineAm = open.ToString("h:mm tt");
Console.WriteLine(nineAm);

TimeOnly close = TimeOnly.Parse("8:00 PM");
string eightPm = close.ToString("h:mm tt");
Console.WriteLine(eightPm);

TimeOnly currentTime = TimeOnly.FromDateTime(DateTime.Now);
string timeNow = currentTime.ToString("HH:mm tt");
Console.WriteLine(timeNow);

if (currentTime < open)
{
    Console.WriteLine("The store has not opened yet.");
}
else if (currentTime >= open && currentTime < close)
{
    Console.WriteLine("The store is open.");
}
else
{
    Console.WriteLine("The store is closed.");
}
//~~~~~
~~~~~

//Part 6 – Formatting + Logic Together
//Exercise 13 – Pharmacy

//Pharmacy:
//Opening time: 08:30 AM

//Closing time: 07:30 PM

```

```

//The program should display:
//Current time: ...
//Opening time: ...
//Closing time: ...

//All three hours must be in the format:
//HH:mm

//Then:
//The pharmacy is open.
//OR:
//The pharmacy is closed.

TimeOnly currentTime = TimeOnly.FromDateTime(DateTime.Now);
string timeNow = currentTime.ToString("HH:mm");
Console.WriteLine($"Current time: {timeNow}");

TimeOnly openingTime = TimeOnly.Parse("08:30 AM");
string eightThirtyAm = openingTime.ToString("HH:mm");
Console.WriteLine($"Opening time: {eightThirtyAm}");

TimeOnly closingTime = TimeOnly.Parse("07:30 PM");
string sevenAndAHalfPm = closingTime.ToString("HH:mm");
Console.WriteLine($"Closing time: {sevenAndAHalfPm}");

if (currentTime >= openingTime && currentTime < closingTime)
{
    Console.WriteLine("The pharmacy is open.");
}
else
{
    Console.WriteLine("The pharmacy is closed.");
}
//~~~~~
~~~~~

```

```

//Final Challenge – TimeOnly Mastery

//A coffee shop is open between:
//06:45 AM
//and:
//09:15 PM

//Create a program that does all of the following:
//Stores the opening time in TimeOnly.
//Stores the closing time in TimeOnly.
//Gets the current time.

//Displays the three hours in HH:mm format.

//Checks if the current time is within the operating hours.
//At exactly the opening time – the coffee shop is open.
//At exactly the closing time – the coffee shop is closed.
//At the end, one of two messages is displayed:
//The coffee shop is currently open.

//Or:
//The coffee shop is currently closed.

TimeOnly currentTime = TimeOnly.FromDateTime(DateTime.Now);
string timeNow = currentTime.ToString("HH:mm");
Console.WriteLine($"Current time: {timeNow}");

TimeOnly open = TimeOnly.Parse("06:45 AM");
string sixtyFourAm = open.ToString("HH:mm");
Console.WriteLine($"Opening time: {sixtyFourAm}");

TimeOnly close = TimeOnly.Parse("09:15 PM");
string nineFifteenPm = close.ToString("HH:mm");
Console.WriteLine($"Closing time: {nineFifteenPm}");

if (currentTime >= open && currentTime < close)
{
    Console.WriteLine("The coffee shop is currently open")
}

```

```
n.");  
}  
else  
{  
    Console.WriteLine("The coffee shop is currently closed.");  
}
```